TITLE – Student Performance Analysis and Prediction System

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Introduction:

This repository contains a **Machine Learning-powered Student Performance Prediction System** built using **Python, Flask, SQLite, and scikit-learn**.
It predicts whether a student will **Pass or Fail** based on subject marks using a trained ML model and integrates this with a full Flask web application.

🧠 Project Overview

This project performs the following:

- Collects student marks in subjects (Tamil, English, Maths, Science, Social)
- Cleans and preprocesses the data
- Trains a machine learning model to predict the student's average score
- Integrates the model with a Flask web app to display predictions in the UI
- Allows teachers to add student records with ML-powered result prediction
- Provides a table with all student records and visual indication of Pass/Fail

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🚀 Features

✓ Add student details
✓ Automatic ML prediction (Pass/Fail)
✓ Store student data in SQLite database
✓ View all student records with color-coded results
✓ Download performance data as Excel
✓ Optional visualizations (charts)

📁 Folder Structure

Student-ML-project/ ├── student_performance/ # Flask application code ├── static/ # CSS & static files ├── templates/ # HTML templates ├── student_performance_model.pkl # Saved ML model ├── train_model.py # Script to train ML model ├── database.db # SQLite database ├── requirements.txt # Python dependencies └── README.md # Project documentation
yaml Copy code

🛠 Technology Stack

Component	Technology
Backend	Python, Flask
Database	SQLite
ML Model	scikit-learn
Frontend	HTML, CSS
Data Processing	pandas
Deployment	Local / Cloud

🧪 **How to Run Locally**

Install dependencies

bash Copy code pip install -r requirements.txt Train the ML model

bash Copy code python train_model.py Run the Flask app

bash Copy code python app.py Open in browser

text Copy code http://127.0.0.1:5000

Software Requirements:

Operating System – Windows 11 Pro, Linux and mac

IDLE – Visual Studio Code

Hardware Requirements: RAM – Minimum 4GB Processor – Minimum intel i3